

Lab 1: GO FOUNDATION

Net-Centric Programming

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Part 1: Variables and Basic types

```
● thanhphatchau@Ants-MacBook-Pro lab1 % go build main.go
● thanhphatchau@Ants-MacBook-Pro lab1 % go run .
=== Property Information ===
Name: Saigon Apartment
PriceL 76 VND
Berrooms: 2
Available: &t
%!(EXTRA bool=true)Price per m^2: 33112583 VND
○ thanhphatchau@Ants-MacBook-Pro lab1 %
```

Successfully installed and run by the computer.

Task 1.1: Add more properties

```
=== Property Comparison ===
Property 1: Saigon Apartment - 33112583 VND/m² | Price: 2500000000 VND | Area: 75.5 m² | Beds
: 2 | Available: Yes
Property 2: Thu Duc Condo - 30645161 VND/m² | Price: 1900000000 VND | Area: 62.0 m² | Beds: 2
| Available: Yes
Property 3: Binh Thanh Studio - 31111111 VND/m² | Price: 1400000000 VND | Area: 45.0 m² | Bed
s: 1 | Available: No

Cheapest per m²: Thu Duc Condo at 30645161 VND/m²
```

Task 1.2: Price Categories

```
Choose option: 2

=== Property Categories ===
Saigon Apartment: PREMIUM
Thu Duc Condo: PREMIUM
Binh Thanh Studio: PREMIUM

Category Summary:
LUXURY: 0 properties
PREMIUM: 3 properties
STANDARD: 0 properties
BUDGET: 0 properties
```

Part 2: Arrays, Slices, and Maps

Task 2.1: Property Search

```
Choose option: 3

--- Search by Budget ---
Enter your max budget (VND): 2000000000
Minimum bedrooms (enter 0 to skip): 2
Only show available properties? (y/n): y

Results:
- Thu Duc Condo | Price: 1900000000 VND | 30645161 VND/m² | Area: 62.0 m² | Beds: 2 |
  Available: Yes
Total matches: 1
```

Task 2.2: District Analysis with Maps

```
=== Property Analyzer Menu ===
1. View all properties and show the cheapest property by price
2. Categorize Property
3. Property Search
4. District Analysis with Maps
0. Exit
Choose option: 4

=== District Analysis ===

[District 1]
Count: 1
Average price: 2500000000 VND
Most expensive: Saigon Apartment (2500000000 VND)

[Thu Duc]
Count: 1
Average price: 1900000000 VND
Most expensive: Thu Duc Condo (1900000000 VND)

[Binh Thanh]
Count: 1
Average price: 1400000000 VND
Most expensive: Binh Thanh Studio (1400000000 VND)
```

Part 3: Functions and Methods

Task 3.1: Investment Calculator

```
Choose option: 5

=== Investment Calculator (ROI) ===
1) Saigon Apartment | Price: 2500000000 VND | District: District 1
2) Thu Duc Condo | Price: 1900000000 VND | District: Thu Duc
3) Binh Thanh Studio | Price: 1400000000 VND | District: Binh Thanh
Choose a property (number): 3
Monthly rent income (VND): 400000000
Annual appreciation rate (%): 2
Annual maintenance cost (VND): 100000000

--- Result ---
Property: Binh Thanh Studio
Annual rent: 480000000 VND
Appreciation gain: 2800000 VND
Annual maintenance: 10000000 VND
Net annual gain: 472800000 VND
ROI: 337.71%
Investment grade: EXCELLENT
```

Task 3.2: Loan Calculator

```
Choose option: 6

=== Loan Calculator ===
Loan amount (VND): 1000000000
Annual interest rate (%): 10
Loan term (years): 10

--- Result ---
Monthly payment: 13215074 VND
Total payment: 1585808843 VND
Total interest: 585808843 VND
```

Part 4: Control Flow and Logic

Task 4.1 Smart Recommendation System

```
=== Smart Recommendation System ===
Your budget (VND): 10000000000
Minimum bedrooms (0 to skip):
Please enter a valid integer.
Minimum bedrooms (0 to skip): 2

Top Recommendations:
1) Saigon Apartment | District: District 1 | Score: 6 | 33112583 VND/m² | Price: 2500
000000 VND | Area: 75.5 m² | Beds: 2 | Available: Yes
2) Thu Duc Condo | District: Thu Duc | Score: 3 | 30645161 VND/m² | Price: 1900000000
VND | Area: 62.0 m² | Beds: 2 | Available: Yes
```

Task 4.2: Property Portfolio Optimizer

```
=== Property Portfolio Optimizer ===
Enter total budget (VND): 10000000000
Expected monthly rent per property (VND): 50000000
Expected annual appreciation rate (%): 3

Selected Portfolio:
1) Binh Thanh Studio | Price: 1400000000 VND | ROI: 45.86% | District: Binh Thanh
2) Thu Duc Condo | Price: 1900000000 VND | ROI: 34.58% | District: Thu Duc
3) Saigon Apartment | Price: 2500000000 VND | ROI: 27.00% | District: District 1

Total spent: 5800000000 VND
Remaining budget: 4200000000 VND
```